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Author(s): Vance H. Fried and Robert D. Hisrich

Source: *Financial Management*, Autumn, 1994, Vol. 23, No. 3, Venture Capital Special Issue (Autumn, 1994), pp. 28-37

Published by: Wiley on behalf of the Financial Management Association International

Stable URL: <https://www.jstor.org/stable/3665619>

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# Toward a Model of Venture Capital Investment Decision Making

Vance H. Fried and Robert D. Hisrich

*Vance H. Fried is Associate Professor of Management at Oklahoma State University, Stillwater, Oklahoma. Robert D. Hisrich is A. Malachi Mixon III Chair in Entrepreneurial Studies, Case Western Reserve University, Cleveland, Ohio.*

**Abstract:** This paper uses a case study methodology to develop a model of the venture capital investment decision-making process. It identifies generic criteria that venture capitalists use. A six-stage process model is proposed: origination, venture capital firm-specific screen, generic screen, first-phase evaluation, second-phase evaluation, and closing. Different activities occur in each stage. It is argued that the venture capitalist provides both supply-side and demand-side benefits to the market.

■ The venture capital market plays a significant role in providing capital to a wide variety of enterprises. While Fried and Hisrich (1994), Sahlman (1990) and Barry, Muscarella, Peavy, and Vetsuypens (1990) have examined the value of a venture capitalist after an investment is made, this paper focuses on the way the venture capitalist makes the investment decision. (In the rest of this paper, venture capitalists will be referred to as VCs.)

The decision to invest is a difficult one with serious adverse selection risk. Once an investment is made, the investment is illiquid, and its success is highly dependent on a small group of managers/entrepreneurs. Significant information asymmetries allow managers to engage in opportunistic behavior after an investment is made (Sahlman (1988)), making it all the more important that the initial decision to invest be a good one.

Opinions vary about how successful VCs are in allocating resources. Chan (1983) and Sahlman (1990) argue that the presence of a VC encourages efficient capital allocation. Amit, Glosten, and Muller (1990a, p.110), though, are more critical, stating:

Under the current institutional structure of the venture capital industry, the most promising entrepreneurs will not seek venture capital financing, and are likely to make slower progress in the development and commercialization of emerging technologies. Further, those entrepreneurs that are backed by venture capital are less likely to

succeed in developing their ventures because of their relatively low ability.

If we are to know whether the venture capital market allocates resources properly, we need to understand how VCs make investment decisions. This problem can be expressed in two related questions: "What criteria do VCs use to evaluate potential investments?" and "What process do VCs use to evaluate potential investments?"

Our study of venture capitalist decision making uses a two-stage research methodology. First, we gathered data from eighteen VCs in three different regions of the United States. Personal interviews about the investment process each used on their most recent investments yielded information that we analyzed for generic criteria. We then developed a model of the investment process. The second stage involved verifying the model and its generic criteria.

There is prior research on the criteria VCs use in evaluating potential investments. Some examples are Tyebjee and Bruno (1984), MacMillan, Siegel, and Subbanarasimha (1985), and MacMillan, Zemmann, and Subbanarasimha (1987). These researchers use relatively simple, structured-mail questionnaires asking VCs to rank the importance of various criteria. This methodology has been criticized on several points by Sandberg, Schweiger, and Hofer (1988).

Studies by Hisrich and Jankowicz (1990) and Hall and Hofer (1993) examine this topic using significantly different methodologies. Neither work examines the entire process involved in making an investment decision. Sandberg et al. (1988, p.13) summarize the problems with the existing research:

Funding support was provided by the Oklahoma Department of Commerce. The authors are grateful to Minu Mathur for her assistance in data analysis and to Garry Bruton and John Polonchek for their comments on earlier drafts of this manuscript.

It is gradually becoming clear that human decision making cannot be understood by simply studying final decisions. The perceptual, emotional and cognitive processes which ultimately lead to the choice of a decision alternative must also be studied if we want to gain an adequate understanding of human decision making.

The earliest VC decision process study we find is a study of VCs in Pittsburgh by Wells (1974). This model of the process is modified and elaborated on by Tyebjee and Bruno (1984). While this model defines stages in the process, the underlying Tyebjee and Bruno study focuses on investment criteria and does not examine the specific activities that VCs undertake.

Our work examines the entire process, using actual cases. From our study of the cases, we propose a model of the process and identify generic investment criteria.

The VC investment decision-making process is designed to reduce the risk of adverse selection. The VC aids suppliers of capital by serving both to produce information and to make decisions. VCs also benefit the users of capital by providing both equity (on a confidential basis) and non-financial resources, such as business advice. By reducing information and contracting costs, the VC can supply equity to firms too small to access the public equity market efficiently.

## I. Methodology

Traditional financial research methodologies are not well-suited to an examination of the venture capital investment process. The transactions involved are private. What little information that is available from secondary sources describes the ultimate investment but not the process used to make it.

Our two-stage methodology addresses this problem. The initial stage is a multiple-case study to gather information on which to base a model. Then, an industry panel critiqued and verified the model.

### A. Initial Study

To obtain data with more informational content than that used in prior studies (Harrigan (1983)), we interviewed 18 VCs personally and followed up with a structured questionnaire. We choose six representative VCs in each of three regions with significant venture capital activity—Silicon Valley, Boston/Cambridge, and the Southwest United States. Choices were based upon recommendations of a panel of industry experts. Each person interviewed is associated with an established venture capital fund. The amount of funds under management varies from \$4 to \$450 million, with a mean fund size of \$99 million and a median fund size of \$50 million.

VCs were asked to describe the investment process each followed in their most recent investment as lead investor. Focus on an actual investment eliminates any problems associated with hypothetical or non-contingent responses. Use of the most recent investments helps to ensure that the recollection is more accurate. Because the financial results of the investment were then unknown, this technique also minimizes any tendency of self-reporting bias. Finally, and perhaps most important, sequential decision making could be examined.

The authors (who have extensive interviewing experience) conducted and recorded all interviews. The transcripts gave us 18 case studies describing the investment process in a variety of different industries and stages of company investment. Table 1 provides a brief description of each case. Investments studied range in size from \$50,000 to \$6 million.

**Table 1. Investments in Study**

| Investment Stage  | Product or Service                               |
|-------------------|--------------------------------------------------|
| Seed              | Specialty Semi-conductors                        |
| Seed              | Artificial Intelligence                          |
| Seed              | Specialized Computer Systems                     |
| First             | Miniaturized Computer Displays                   |
| First             | Video Surveillance Systems                       |
| First             | Educational Software                             |
| First             | Software for Microwave Integrated Circuit Design |
| First             | Specialty Materials for Semi-Conductors          |
| First             | Surgical Laser                                   |
| First             | Engineering Software                             |
| Second            | CAD/CAM Systems                                  |
| Second            | Telecommunication Services                       |
| Second            | Computer Systems for Handling Parcels            |
| Second            | Commercial Beef                                  |
| Management Buyout | Surgical Laser                                   |
| Management Buyout | Data Entry Terminals                             |
| Leveraged Buyout  | Mining Pumps                                     |
| Recapitalization  | Carpet                                           |

The VCs were asked to complete a questionnaire about the amount of time involved in the process and how often they undertake activities such as reference checking or commissioning a formal marketing study. The questionnaire also asked for descriptive data about the firm, such as fund size and investment stages of particular interest.

One of the authors and an outside researcher independently tried to fit each case into the Tyebjee and Bruno model. While the results were generally supportive of that model, there were some major exceptions. A modified model was then created using a replication logic (Eisenhardt (1989)). This approach proposes modifications to a model to deal with flaws in the initial version. Modifications are then tested against each case until the case supports the proposed modification. The resulting model was then verified through a second study (discussed later).

The cases and written questionnaires were analyzed for specific activities performed by venture capitalists. Because the characteristics of a venture vary by stage of development (Kazanjian and Drazin (1989), Ruhnka and Young, (1987)), the VCs were divided into late- and early-stage investors.

In the first two interviews, the VCs were asked how they assessed the various investment criteria suggested by MacMillan et al. (1985). Responses indicated that these VCs do not use these explicit criteria in evaluating ventures. These comments, coupled with the numerous objections raised by Sandberg et al. (1988) to the MacMillan criteria indicate that the MacMillan criteria do not adequately describe the VCs' criteria. We then used a grounded theory approach to develop a new set of generic criteria (Glaser and Strauss (1967)).

The grounded theory approach is a highly iterative process that uses cross-case comparisons to develop hypotheses. The hypotheses are then tested against each case. Hypotheses are then refined and new hypotheses developed on the basis of these tests and results from earlier published studies. The resulting generic criteria were then examined by an industry panel.

## B. Industry Panel

After development of the model and generic criteria, we presented the results of the initial study to five new VCs, selected to represent a broad range of firms and geographic areas. Two of the five are large Midwestern firms emphasizing late-stage investments; one is a large Silicon Valley/New York firm that invests in all stages; one is a New York venture capital subsidiary of a major European bank that focuses on large, late-stage investments; and one is a moderately small, Pittsburgh-based VC firm that invests in all stages.

Each was mailed a summary of the model and generic criteria. They also received a copy of Table 2 with brief comments on the differences observed between early- and late-stage investors. We made further modifications to the model based on their responses.

## II. Results

We describe results first in terms of the generic criteria and then as they fit the process model.

### A. Generic Criteria

Fifteen criteria common to the investments studied were identified. Given the breadth of investments studied, it seems safe to call these criteria a common base for VC investments. These criteria expand upon the three basic constructs identified by Hisrich and Jankowicz (1990): concept, management, and returns. *Concept* has four components. First, there must be significant potential for earnings growth. This is obvious for early-stage ventures, which may not be generating revenues yet, but it is also important for late-stage investments. It is difficult, if not impossible, to get venture capital rates of return (30% to 70% IRRs) without significant earnings growth. Earnings growth may come from a rapidly growing market or through increasing market share or significant cost cutting.

Second, the investment must involve a business idea (new product, service, or retail concept) that works already or can be brought to market within two to three years. Third, the concept must offer a substantial "competitive advantage" or be in a relatively non-competitive industry. Fourth, the concept must have reasonable overall capital requirements. This criterion was added at the suggestion of the two large investors in the industry panel. One states:

In today's environment an entrepreneur would be well served to develop a business plan for which the total cash requirement to achieve self-funding is within the reach of the venture capitalist and his original investment group. If not, both he and the original investment group are subject to being washed out in subsequent financings, even if the company has done moderately well, due to the limited availability of growth capital.

The other raised a concern as to whether additional money was available anywhere and also pointed out that high capital requirements could seriously impact ROI.

*Management* is also an important consideration. There are several attributes VCs want to see in managers. First, managers must display personal integrity. Second, they must have done well at prior jobs. The track record does not have to be with the current company. Association with losing ventures in the past does not disqualify the entrepreneurs if they can show they personally performed well in the earlier venture. In later-stage investments, the focus tends to be more on performance on the current job.

Managers must be realistic. VCs try to judge the

manager's ability to identify risk and, where appropriate, develop plans for dealing with these risks. Managers also need to be hardworking, flexible, and have a thorough understanding of the business. Flexibility is especially important for early-stage ventures.

Managers must exhibit leadership, not only in good times but also under extreme pressure. Finally, managers must have general management experience. Leadership capabilities and general management experience may not be as significant in an early-stage venture as long as the entrepreneur is willing to add additional management, possibly at the CEO level, to correct this deficiency.

*Returns* has three components. First, the investment must provide an exit opportunity. VCs generally reap returns by a public offering, sale of the company, or buyback of the VC investment by the company. VCs do not expect easy liquidity, but they require the likelihood of some type of exit in a three- to ten-year period. Second, the investment must offer the potential for a high rate of return. Hurdle internal rates of return are in the 30-70% range. Finally, the venture must offer the potential for a high absolute return. VCs are not willing to spend time on small investments that offer low absolute returns, even if the rate of return is high.

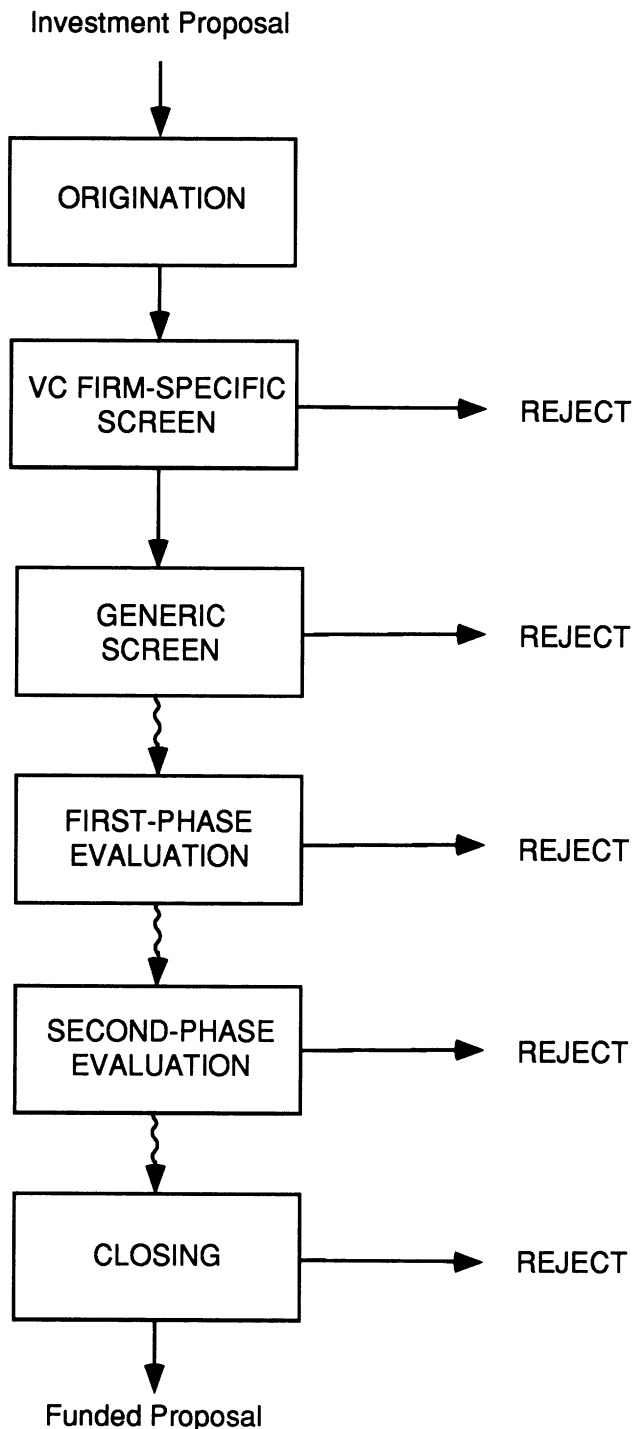
These are broad generic criteria. The specifics of each criterion may vary from VC to VC. Opinions will differ on acceptable rates of return as well as the importance of various criteria. For example, some VCs will invest in a strong concept even if they view management as a question mark. Others will invest only if they feel management is very strong. Even if two VCs have the same criteria, there may be major differences in their judgment as to how well a particular investment proposal meets these criteria.

## B. A Model of the Decision-Making Process

The decision-making process can be modeled in six stages as shown in Figure 1. The process is time-consuming and labor-intensive. VCs indicate it takes on average 97.1 days (s.d. = 45.0) for an investment to pass through these six stages and receive funding. They estimate that on an average funded proposal for which they serve as lead investor, the process requires 129.5 hours of their time (s.d. = 99.8).

**Origination.** The first phase is origination. While VCs generally wait for deals to come to them, they do make themselves known to companies through industry directories. The bulk of the VCs' efforts to generate investment proposals focus on developing a network of referrers. While VCs receive many deals "cold" (without any introduction), they rarely invest in them. None of the 18 deals in the study was received cold. Most funded proposals come by referral. Occasionally, the VC already knows the

Figure 1. Venture Capital Investment Process



founder through work either as a manager of a prior investment or a consultant.

The referred deals in our study come from a variety of sources: investment bankers, investors in the VCs' fund, commercial bankers, management of firms in the VCs' portfolio, consultants who had worked for the VC in the past, and family friends. There are two reasons for this heavy incidence of referrals. First, referred deals are more likely to pass through the generic screen if the VC has confidence in the referrer's judgment. Second, the referrer is more likely to understand what type of investments the VC might find attractive.

A growing minority of VCs aggressively seek out deals and, in some cases, may actually help create them. In one case in the study, the VC met the entrepreneur at a party when the entrepreneur was working for a *Fortune* 100 company. When they discussed a technology in which the VC was very interested, the VC was impressed by the entrepreneur's knowledge and invited the entrepreneur to contact him if he was ever interested in starting his own company. A year and a half later the entrepreneur called—without a business plan and with only a rudimentary knowledge of business. The VC worked with the entrepreneur to develop the business side of the concept and to write a business plan. Only after the plan was written did the VC invest any money.

**VC Firm-Specific Screen.** Many VCs have firm-specific criteria on investment size, industries in which they invest, geographic location of the investment, and stage of financing. The firm-specific screen eliminates proposals that clearly do not meet these criteria.<sup>1</sup> At most, the firm specific-screen involves a cursory glance at the business plan without any analysis of the proposal.

**Generic Screen.** Many proposals pass through the firm-specific screen only to be rejected without extensive review when the VC analyzes the proposed investment in terms of the generic criteria. Most deals that pass through the firm-specific screen are rejected at the generic screen based upon a reading of the business plan coupled with any existing knowledge the VC may have relevant to the proposal. The generic screen will be less rigorously applied when the quality of referrer is high. The net result of the two screens is that most proposals are rejected with minimal investment of time.

**First-Phase Evaluation.** After proposals pass through the generic screen, the VC begins to gather additional information about the proposal. One VC with experience as

a commercial lender in a money center bank told us that the VC process is much more involved than bank loan reviews. At the same time, there are of course some limits. As one VC says: "If you really wanted to have *all* the market checks and talk to *all* the customers prior to putting in your first dollar, not very many ventures would get off the ground."

Table 2 provides a list of activities that VCs undertake in the evaluation phases. Evaluation involves general monitoring by the VC. In these phases, the information gathered from both company and outside sources is compared to the information in the entrepreneur's business plan. "We filter all these reports in just like an intelligence officer would in the army—a lot of sources—you know, some reports are good and some reports are bad."

After clearing the generic screen, a proposal's progress through the remaining stages is not preordained, as is indicated by the wavy line in Figure 1. As one VC comments:

It's not a nice slow curve where you sort of get mildly interested and then come to a crescendo at the end. It's more the other way around. And you have peaks and valleys in the middle.

The first-phase evaluation generally starts with a meeting with the principals of the company seeking financing. As the proposal is being evaluated, a series of meetings with all the top management team will occur. These meetings have two goals: to increase the VC's understanding of the business and to allow assessment of the manager's understanding of the industry, the proposal, and problems that may be encountered. They also provide an opportunity to assess how managers think and behave.

One VC even wants to visit an entrepreneur's home and meet the family. He explained:

I like to go out and meet the spouse and the kids and try to see if their home is in chaos or if it's pretty orderly. Rather than me passing judgment on how they live, what I want to understand is what kind of environment that entrepreneur's coming out of.

Evaluation meetings provide an opportunity to assess the ability of management to react under pressure. As one VC states:

We subject management to a number of sessions where we might lean on them a little bit to see what kind of pressure points they have.... We see how they interact with their management team, and we ask questions where we know there's going to be some controversy to see how they handle it.

Management's abilities are also assessed by checking references provided, as well as others not identified by the

<sup>1</sup> A VC on the industry panel comments, "While most venture capital firms profess to have a firm-specific screen, they are opportunistic, and investment opportunities outside that screen may still have an opportunity to be funded."

**Table 2. Evaluation Activities Carried Out by Venture Capitalists**

Venture capitalists were asked: "Prior to funding an investment as lead investor, how often do you engage in the following activities?"

| ACTIVITY                                                                  | HOW OFTEN (%) |
|---------------------------------------------------------------------------|---------------|
| Interview all members of management team                                  | 100           |
| Tour facilities                                                           | 100           |
| Contact entrepreneur's former business associates                         | 96            |
| Contact existing outside investors                                        | 96            |
| Contact current customers                                                 | 93            |
| Contact potential customers                                               | 90            |
| Investigate market value of comparable companies                          | 86            |
| Have informal discussions with experts about the product                  | 84            |
| Conduct in-depth review of pro forma financials prepared by company       | 84            |
| Contact competitors                                                       | 71            |
| Contact banker                                                            | 62            |
| Solicit the opinion of managers of some of your other portfolio companies | 56            |
| Contact suppliers                                                         | 53            |
| Solicit the opinion of other venture capital firms                        | 52            |
| Contact accountant                                                        | 47            |
| Contact attorney                                                          | 44            |
| Conduct in-depth library research                                         | 40            |
| Secure formal technical study of product                                  | 36            |
| Secure formal market research study                                       | 31            |

entrepreneur. The extent of reference checking varies depending on the VC's prior knowledge of the entrepreneur. According to one VC, the ability to get references, as well as other people, to "really talk" is one of the factors differentiating a good VC from a bad one.

Late-stage investors are more likely to talk to accountants than early-stage investors, reflecting applicability of the financial history available. Information availability also leads late-stage investors to engage more frequently in library research. Late-stage investors are also more likely to talk to banks, but banks are not viewed as particularly frank sources, especially if owed money.

Both existing and potential customers are contacted to determine why they are buying or not buying from the company. In the case of the educational software company, the VC accompanied the company's management to sales presentations made to school boards to see the reaction of potential customers. Similarly, early-stage investors may contact potential customers before a product has been fully developed. In the case of the semi-conductor materials producer, the VC, through a personal contact, located a packaging engineer at a major chip company that would be

a potential purchaser of the product and discussed the product concept.

Formal market studies may also be made, sometimes by outside consultants. VCs report doing this only about 30% of the time, for several reasons. First, a great deal of information is almost always in the business plan. Second, contacts with customers and potential customers provide additional information. Third, sometimes the market is not clearly defined.

In the case of the microcomputer display, the VC felt that the market was not totally analyzable, as nothing similar to the product existed. He explained:

Basically you look at the applications—radio pagers. How many radio pagers are sold per year? How many of those radio pagers are alpha-numeric versus simple-minded numeric ones?... So you got a gazillion radio pagers.... And you factor that out and you say, "OK, this will get 10 or 15% of the radio pager market." And you can work through some numbers, but in the end you have to...say that people will need this display, have to put up with some idiosyncrasy.

Technical studies of a product are used much more by early-stage investors than late-stage, because late-stage investors can get a good feel for the state of the company's technology by talking with customers and industry experts. Early-stage investors achieve their technological evaluation in a variety of ways. Several early-stage investors have formal affiliations with technology experts. One of the firms in our study is associated with a consulting firm well-known for technology assessment. Another VC firm uses "technology partners," consultants formally affiliated with the firm who draw a retainer and participate in profits but do not make any investment decisions. Other early-stage investors handle technology assessment on an informal, ad hoc basis.

VCs often discuss an investment with managers in some of their existing portfolio companies, particularly with those in closely-related industries. These portfolio companies might also be customers or potential customers, or suppliers or potential suppliers, of the company under consideration. Early-stage investors tend to contact management of portfolio companies more than late-stage investors because they tend to specialize more by industry and are therefore more likely to have portfolio company managers with relevant information.

VCs also talk to each other. Their experience with proposals they have analyzed and investments they have made gives VCs knowledge that might be useful to others. In fact, as Reiner's history of the industry (1989) points out, VCs have traditionally invested through loose syndicates. To some extent this is to pool capital in order to share risk and increase the absolute amount of capital that can be invested in any one company. Syndicates are also formed in order to share knowledge, according to Bygrave and Timmons (1992). In fact, some of the firms in our study will not become the sole investor in a company. (Lerner (1994) describes the syndication process.)

VCs analyze pro forma financial projections prepared by the entrepreneur to assess a project's potential for earnings growth, as well as gain information about management's understanding of their proposal and their realism toward its future. The financial projections provide a basis for comparison with the market value of other companies, to give the VC an estimate as to the potential value that can be received when it exits the investment. (See Morris (1989) for an example of how valuation is done by a late-stage investor.) Late-stage investments have more historical financial information available, which allows a more meaningful analysis.

**Second-Phase Evaluation.** At some point the VC develops an "emotional" commitment to a proposal, which

marks the start of the second phase of the evaluation process. Evaluation activities continue, but the amount of VC time spent on the proposal increases dramatically, and the VC's goal changes. While in the first phase the goal is to determine whether there is serious interest in a deal, in the second phase the object is to determine what the obstacles to the investment are and how they can be overcome.<sup>2</sup>

The degree to which firms formally recognize the movement from first to second evaluation phase varies greatly. The same is true for the shift from second-phase evaluation into the closing stage. These last three stages are not clearly distinguishable in all VC firms.

Because of the significant amount of time spent in the second phase, VCs like to have at least a rough understanding about the structure of the deal, including price, *before* entering this phase. This keeps the VC from devoting significant time evaluating proposals that ultimately will not be investable because they are priced too high. Since totally unrealistic initial pricing proposals from the entrepreneur may lead to a quick turn-down, some entrepreneurs submit proposals without a price to avoid negatively influencing the VC in the generic screen and first-phase evaluation stages.

Two of the deals in our study did not require a defined evaluation phase after the formal request for funding was made. In one case, the VC had spent a substantial amount of time helping the entrepreneurs prepare the business plan. With this active involvement in creation of the deal, there was no need for a separate evaluation phase. In the other case, the VC had turned the deal down a few years earlier but had stayed in regular contact with the company since.

**Closing.** After progressing through the second-phase evaluation, the proposal enters the closing stage, where the details of the structure are finalized and legal documents negotiated. After the documents are signed, a check is given to the company. Even though both VC and entrepreneur have invested large amounts of time to get to this final stage, a surprising number of deals that reach this stage are not funded. Some VCs in our study think the percentage is about 20%.

**Proposals with Serious Problems.** As Figure 1 indicates, a proposal can be rejected at any stage in the process. The VC has four options when faced with a problem in the proposal. First, the deal may be further investigated. Two of the cases in the study were almost rejected at the generic screen, but the VC chose to investigate further and ultimately accepted the proposal.

<sup>2</sup>One of the large investors in the industry panel proposes a third evaluation stage. He feels that at some point a VC starts getting serious about a deal, but does not really identify it as something he/she wants to do. This middle stage then ends when the VC either becomes emotionally committed to the deal or loses interest.

Second, the VC can ask for significant changes in the original proposal. This happened in 4 of the 18 cases studied. It is not unusual for new managers to be added as a condition of funding. Third, as was the case twice in this study, the VC may simply go ahead and do the deal even though some serious concerns are present.

Finally, by far the most common response is to reject the deal. Sometimes proposals rejected earlier may reappear in a different form and be accepted. Of the 18 funded proposals in the study, three had initially been rejected by the same VC who ultimately invested.

### III. Discussion

The discussion compares our results to those of prior studies on the investment decision-making process, discusses the benefits provided by the VC intermediary, and suggests areas for further research.

#### A. Comparison with Previous Research

The generic evaluation criteria we develop expand on the three basic constructs identified by Hisrich and Jankowicz (1990). While there are many similarities between these criteria and those proposed by MacMillan et al. (1985), our criteria are more clearly defined and are not limited to new, technology-oriented investments. For example, MacMillan speaks of "proprietary products," a concern for investors in new technology, while we more broadly discuss a business concept offering a "competitive advantage." The differences between our criteria and MacMillan's are not surprising, because MacMillan's aim is to evaluate the importance of various potential criteria to individual VCs; while our purpose is to identify the generic criteria that all VCs use.

While the process model proposed has many similarities with Tyebjee and Bruno's, there are also major differences. The screen phase described by Tyebjee and Bruno is what we call the VC firm-specific screen. When we discovered that most proposals passing through this screen are still quickly rejected, we added the generic screen.

Tyebjee and Bruno do not divide the evaluation phase into two parts. Rather they show only one phase, with pricing negotiated after that. Our model indicates that pricing is negotiated much earlier in the process, with the level of VC evaluation activities increasing significantly after pricing is settled. Our model also points out the significant need for information-gathering activities.

Amit et al. (1990a, 1990b) take a negative view of the venture capital investment process. Their model predicts that the market will break down because of adverse selection. This conclusion is based on the assumption that the

entrepreneur has the options of either investing personal funds or borrowing rather than using venture capital. When this assumption does not hold, their model predicts that the market will not break down. Thus, it is important to determine whether most entrepreneurs seeking VC funding have the option of funding their ventures with personal funds or borrowing.

While our study is not specifically designed to answer this question, as part of the study we did look at 18 entrepreneurs seeking VC financing. None of the 18 had the financial ability to fund their enterprise fully with personal funds or borrowing. Thus under Amit's model, the market should not break down in any of the cases examined.

#### B. Benefits Provided by the VC

Our study indicates that VCs provide benefits to both the supply and demand sides of the market. We find that VCs negotiate basic contract terms fairly early in the process. Sahlman (1988, 1990) claims the contracts VCs negotiate provide benefits because they address many of the moral hazard problems that will arise after the investment. The contracts also serve as a sorting mechanism by allocating much of the risk of failure to the entrepreneur, thereby discouraging entrepreneurs who are not confident of their venture's success.

The VC investment decision-making process reduces the adverse selection problem in several other ways. At origination, a referral network lowers risk, because the referrer, who is generally familiar with the proposal, will want to maintain his/her relationship and standing with the VC.<sup>3</sup> The generic screen further eliminates low-quality proposals. Finally, the few proposals that make it to the evaluation phases are subject to extensive scrutiny and information gathering.

Chan (1983) and Sahlman (1990) both argue that VCs play a major role by serving as producers of information. Our study supports this view. The VC operates in a market with imperfect information and expends a great deal of effort collecting information, usually at less cost than investors would incur gathering the information directly.

The VC has lower information-gathering costs for three reasons. First, the VC is gathering the information on behalf of a number of investors in a limited partnership, giving the VC economies of scale. Second, because the VC invests in different ventures, there are economies of scope, particularly in creating a network of referral sources, service providers

<sup>3</sup>Instances when the VC "creates" the deal do not fit into an adverse selection model.

(attorneys, accountants, management recruiters), and industry contacts (Sahlman (1990)). Third, the VC benefits from a learning curve (Hall and Hofer (1993) and Sahlman (1990)). Information produced for one proposal is useful for subsequent proposals, radically reducing information-gathering costs, especially for the many VCs who specialize by industry.

The VC does not serve merely as an information-producing agent for the investor. The VC is also a decision-making agent. As Sahlman points out, VC investment decisions involve a great deal of uncertainty. Often two investors may come to different conclusions on a project, even with the same information. Most VC partnerships are structured so that the VC has total decision-making authority (*Terms and Conditions of Venture Capital Partnerships* (1987)). In this study, few VCs even consulted with their investors before making an investment.

VCs are better able to make investment decisions than are their investors. Barry et al. (1990) and Megginson and Weiss (1991) show that the capital markets attach value to the post-investment activities of a VC, assuming that VCs are better at overseeing and guiding new enterprises than the investor. Similarly, Fried and Hisrich (1989, 1992) find that venture capital investors want their funds managed by VCs who are experienced, proven decision-makers.

The VC also offers demand-side benefits that may make it more attractive to the entrepreneur than a public equity offering. In addition to providing capital, the VC may provide business advice, contacts, and reputational capital that are not available from the public equity markets (1990).

Another demand-side benefit is confidentiality. A public offering requires the entrepreneur to make certain information public. Additional information may also have to be made public in order to gain market acceptance for a new issue. Much of this information may be proprietary, and perhaps a source of potential economic rents to the

entrepreneur. Public disclosure of this important information may be quite costly to the firm. Campbell (1979) discusses the significance of confidentiality in debt markets.

Finally, Fama and Jensen (1983a, 1983b) point out that public equity is unavailable to most entrepreneurs. The amount of money needed may not be enough to achieve minimum efficient scale in the public equity market. Public equity has high fixed information and contracting costs. By reducing information and contracting costs, the VC is able to provide a source of outside equity to the entrepreneur. To many entrepreneurs, VCs represent not only the best, but also the only source of equity.

### C. Suggestions for Future Research

There are several important areas for future research. First, because the number of VC firms involved in this study is small, the model needs to be validated further on a larger sample through a structured mail questionnaire.

Second, our generic model was developed to cover all types of VCs. Not all VCs are the same, so the differences between various VCs need to be explored. Some differences involve those between early- and late-stage investors. Elango, Fried, Hisrich, and Polonchek (1994) identify other sources of differences, including geographic location and firm size. Also, our study follows only American VC funds. The venture capital industry outside the U.S. has grown explosively in recent years and is now larger than the U.S. industry. Bygrave and Timmons (1992) point out that the nature of the industry varies from country to country.

Finally, one must ask, "How do the benefits that venture capitalists provide as intermediaries compare to the costs?" While we show that the role of an intermediary is valuable, the VC style of investing has high costs. VC funds traditionally pay a management fee of 2.5% of assets per year, as well as 20% of any profits. More empirical work will be needed to explore in what situations venture capital costs are justified. ■

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